

INSTALLING AND SLICING ON CURA FOR ESSENTIUM HSE



Essentium Headquarters

19025 N. Heatherwilde Boulevard

Suite 100

Pflugerville, TX 78660

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Table of Contents

Overview	3
Purpose	3
Intended Audience	3
Platform.....	3
Installing and Slicing – Windows PC	3
Installing and Slicing – Mac Computer	14
Support And Hotend/Nozzle Compatability	26
Important Notes	28

OVERVIEW

Cura is an open source slicing application for 3D printers. This document shows the steps for installing and slicing with the UltiMaker Cura® application using either a Windows® PC or a Mac® computer.

Purpose

The purpose of this Standard Operating Procedure (SOP) is to show the steps for installing and slicing on the UltiMaker Cura application.

Intended Audience

This SOP is intended for any user of an Essentium HSE Printer that needs to leverage the UltiMaker Cura application for slicing 3D parts.

Platform

- Essentium HSE 280i
- Essentium HSE 180

CRITICAL STEPS

Following all of the outlined steps and instructions is critical for proper installation. However, please pay special attention to steps and instructions with the following icon:



Please review the IMPORTANT NOTES section at the end of this document for important usage notes, known issues, and common errors.

INSTALLING AND SLICING – WINDOWS PC

This section shows the steps for installing and slicing Cura using a Windows PC. The following section of this SOP shows these steps, but for a user using a Mac computer.

1. Click on the [Win64.exe](#) link to download Cura v5.6.0.
2. Double click on .exe file, and then follow the Setup instructions in the installer (See Figure 1).



Figure 1: Following the Setup Instructions

3. Open and then immediately close UltiMaker Cura once the software installation is complete (See Figure 2).

NOTE: Opening and immediately closing the software is necessary for performing future steps.

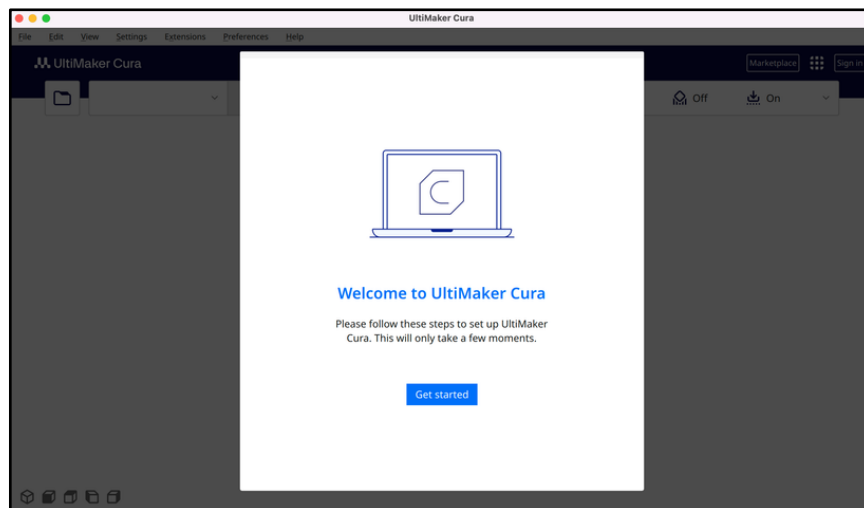


Figure 2: Opening and Then Immediately Closing UltiMaker Cura

4. Download the accompanying Essentium resources files (v1.1.2 zip file), and then unzip (extract) the file. With the zip file is selected, unzip files by navigating to the Extract tab, which is visible along the File Explorer top bar. Select "Extract All".
5. Keep File Explorer for this unzipped folder open on desktop.

6. Navigate to the Cura Configuration folder in File Explorer

- C: > Users > (username) > AppData (hidden) > Roaming > Cura > 5.6
- **NOTE 1:** {user_name} will be specific to each user
- **NOTE 2:** To reveal the AppData folder, Select the 'View' tab on the top bar in File Explorer, and then check the 'Hidden Items' box.

7. Copy and paste the Essentium resources files into the Cura Configuration folder "5.6"

(See Figure 3). This folder will include the following subfolders:

- Definitions
- Extruders
- Intents (Currently empty placeholder)
- Materials
- Meshes
- Quality
- Scripts
- Setting_visibility
- Variants

There are 2 options for migrating the files from the downloaded .ZIP into the Cura Configuration folder. The first option may only be used if you have not saved any custom Cura profiles based on our Essentium materials. This includes saved profiles for all platforms (Essentium or other).

NOTE: If you bulk replace these folders and have custom saved profiles within Cura, you risk deleting your custom saved profiles.

Option 1 is simply selecting all the folders together and drag and dropping the folders into the 5.6 folder, as seen below:

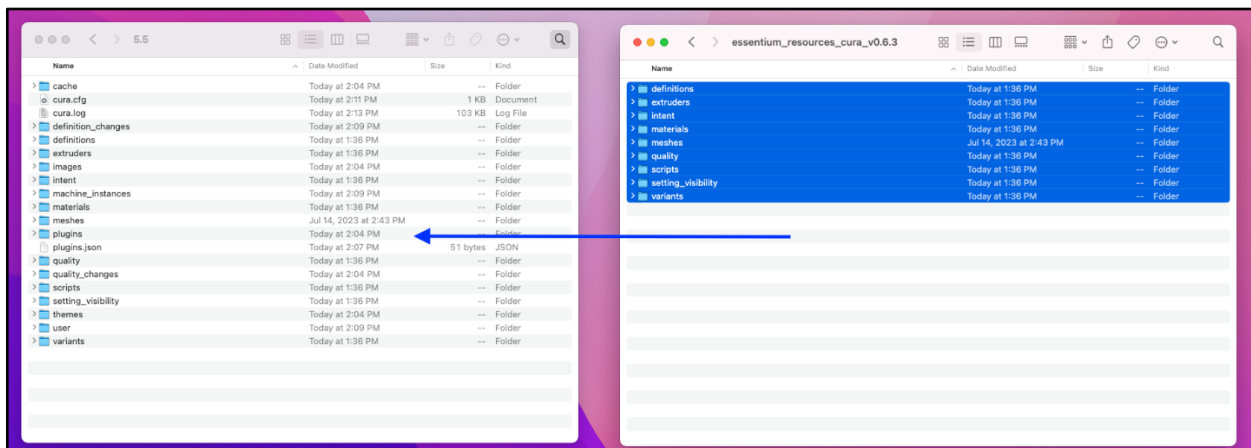


Figure 3: Copying and Pasting the Essentium Resource Files

Option 2 requires opening each individual folder above and dragging and dropping the contents directly into the corresponding folder within the 5.6 folder. This will need to happen for each

folder listed. This ensures that the files in this folder structure related to your custom saved profiles are not harmed during the file migration process.

8. Double-click on the UltiMaker Cura application to re-open Cura. At this point, all Essentium machine and materials are available for selection and use.
9. The 'Getting Started' panel prompts the user to add the first machine into UltiMaker Cura (See Figure 4).

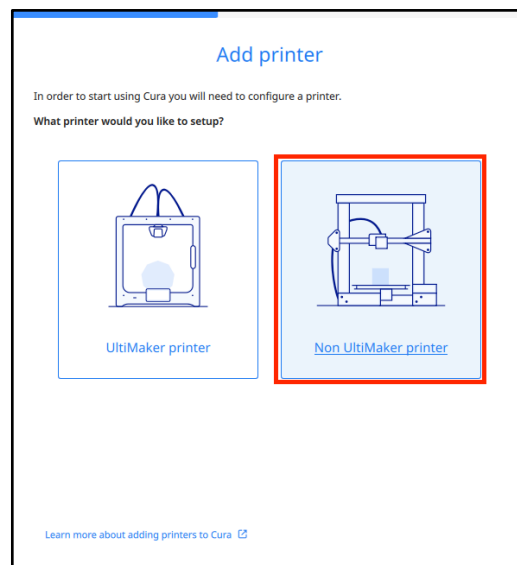


Figure 4: Adding a Printer

Essentium machines are listed under “Essentium, Inc” in the “non-networked” sub-section.

10. Select “Add a non-networked printer”
11. Select the desired HSE machine, and then select ‘Next’ (See Figure 5).

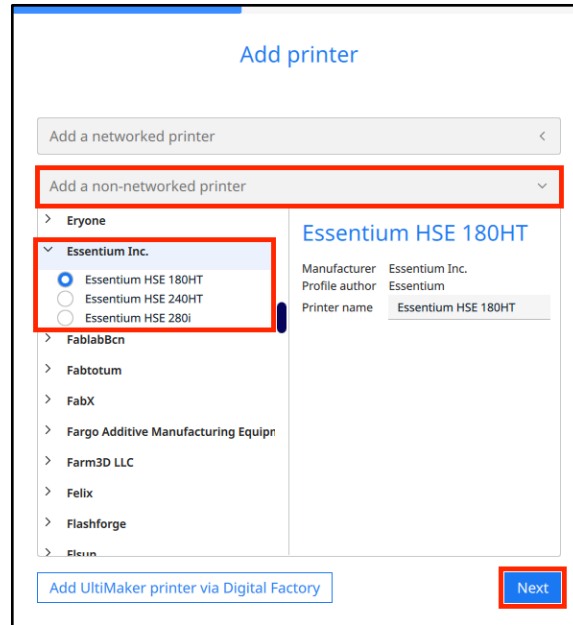


Figure 5: Adding an HSE Printer

12. Enter any desired settings on the 'Machine Settings' prompt, and then click 'Next' (See Figure 6).

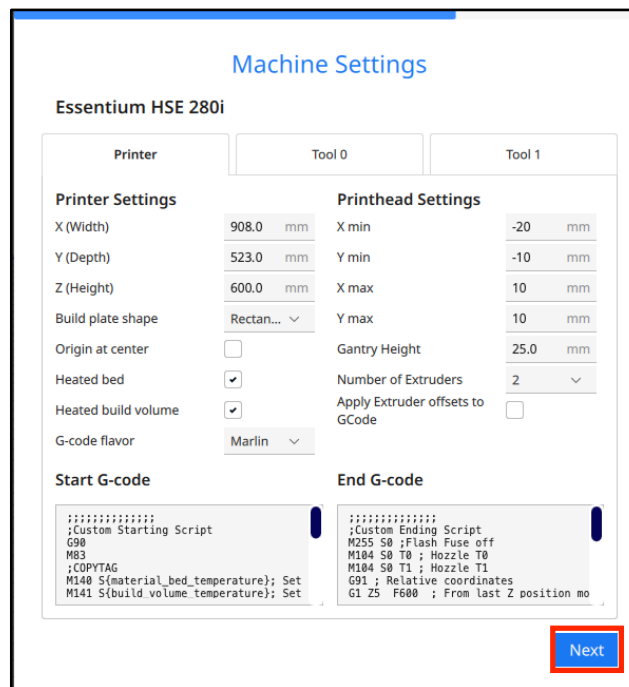


Figure 6: Adding Machine Settings

13. Figure 7 shows where additional machines can be added at the upper-left corner of the screen.

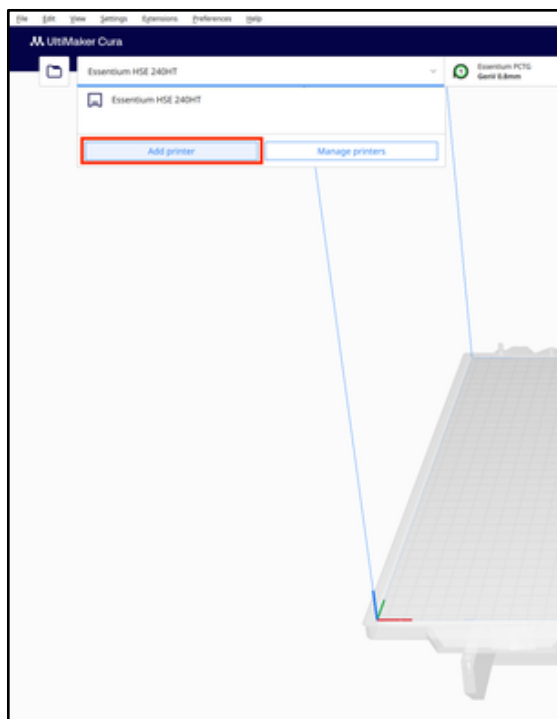


Figure 7: Adding Additional Printers

14. Navigate to Extensions > Post Processing > Modify G-Code (See Figure 8).

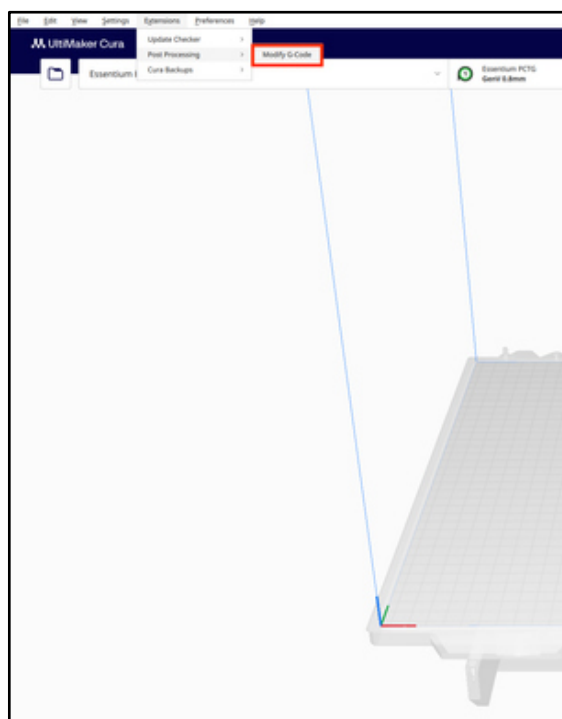


Figure 8: Navigating to Post-Processing Scripts

15. Click on 'Add a script' (See Figure 9).

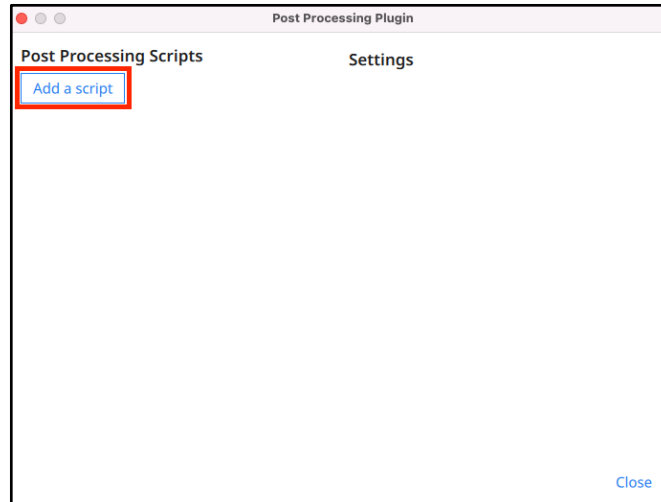


Figure 9: Adding a Script

16. Add the following post-processing scripts by selecting the name from the script, and then clicking 'Close' (See Figure 10):

- Essentium Post Processing Script
- Essentium Setting Injector



NOTE: This step must be done for each machine added.

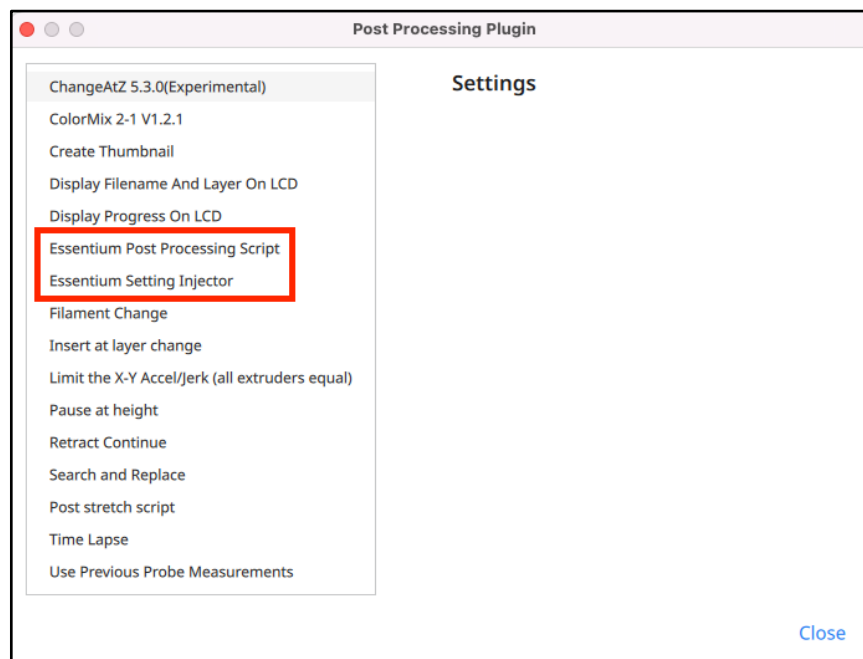


Figure 10: Adding Post-Processing Scripts

17. Select 'Marketplace' at the top-right of the screen, and then search for each plugin name, and then click 'Install' (See Figure 11).



Required plugin:

Linear Advance Settings

Optional plugins:

- Material Settings
- Printer Settings
- Settings Guide

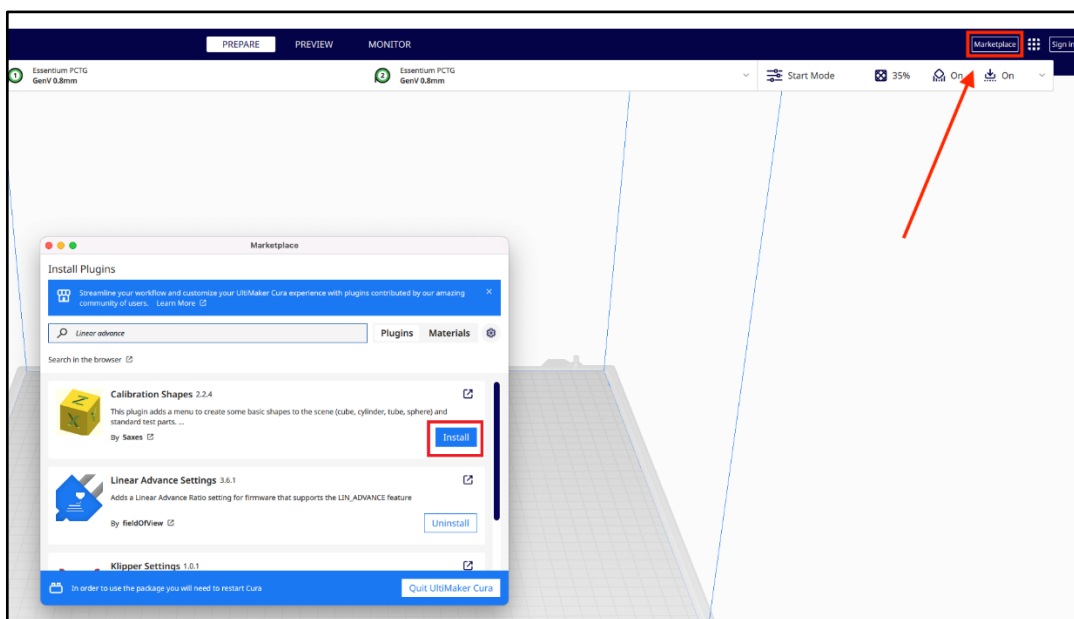


Figure 11: Adding Marketplace Plugins

18. Restart the Ultimaker Cura software for plugins to take effect.

At this point, the Ultimaker Cura software is ready for printing. The steps below show a quick overview for importing a part, *.stl* (or *.obj*), and for selecting the user's materials and hotend/nozzle of choice.

19. Select the desired material and hotend/nozzle (See Figure 12).

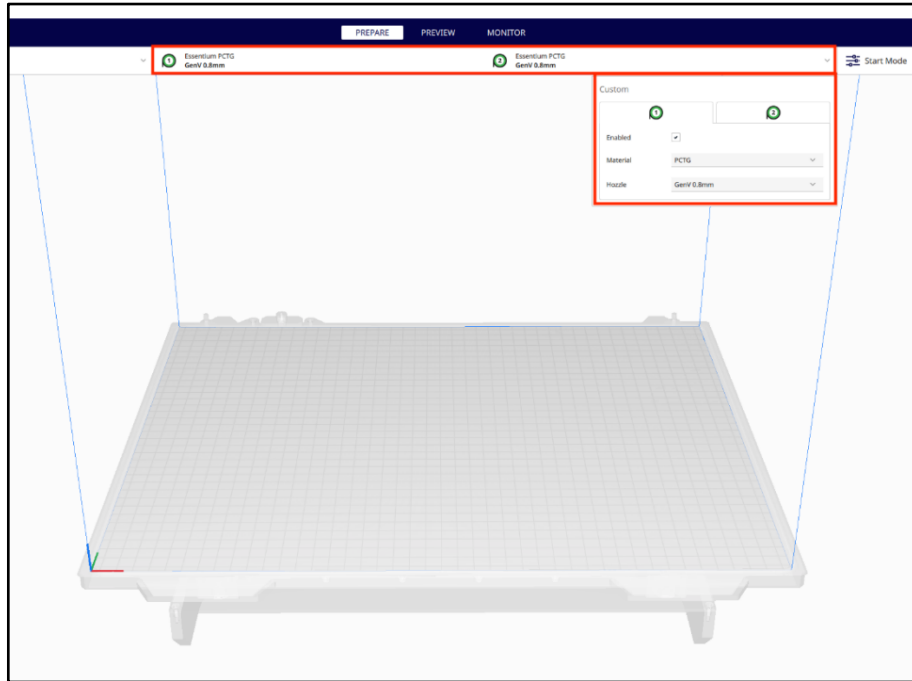


Figure 12: Selecting Material and Hotend/Nozzle

NOTE: At this stage, a majority of the application-specific profiles are not different from “Start Mode”.

20. Select settings visibility. Essentium has a custom option for which settings are visible at the right-hand panel, showing the settings that are most often modified in Cura for HSE printers (See Figure 13).

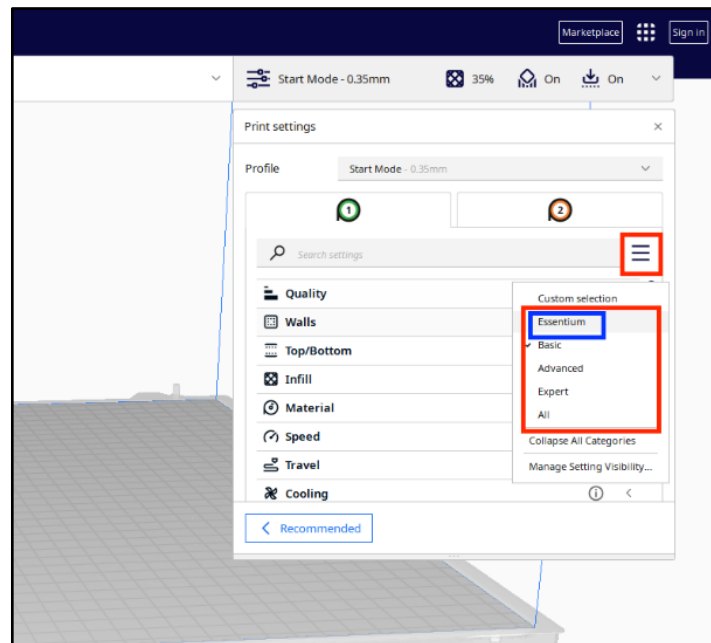


Figure 13: Modifying Settings

21. Click the folder icon in the upper-left corner of the page to select a part (See Figure 14).

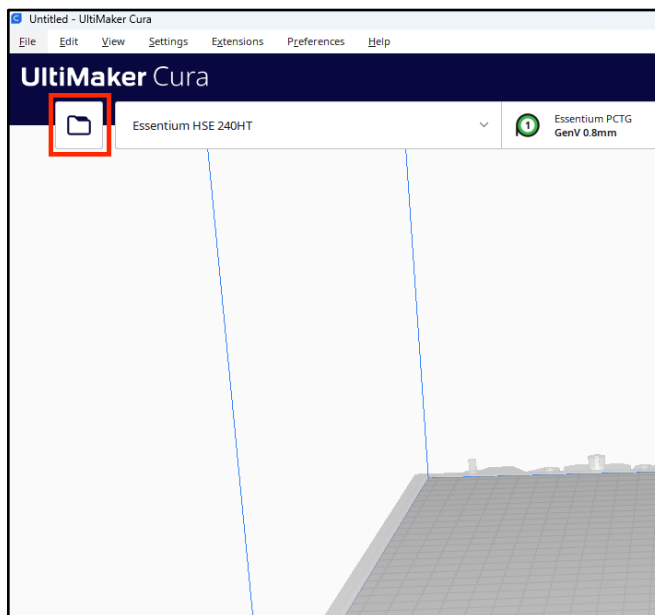


Figure 14: Selecting a Part

22. Click on 'Slice' in lower-right hand corner of the page to begin slicing (See Figure 15).



NOTE: Upon adding a part to the environment, an icon with a red “2” appears next to the ‘Slice’ button. This is an indication that the two (2) post-processing scripts enabled earlier in this procedure are active. Verify that this icon appears next to the ‘Slice’ button when printing on HSE machines, as the print will not execute without these post-processing scripts enabled.

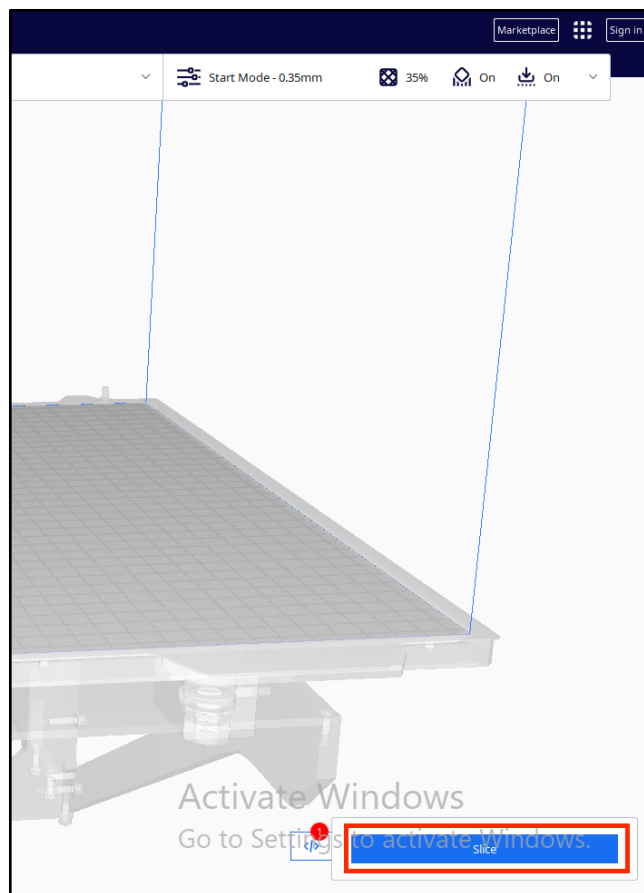


Figure 15: Slicing

At this point, the user can view the toolpath created for that specific part as well as save the .gcode.

23. (Optional Step) The 'Save' button only shows up once a modification is made from the base settings.

NOTE: These user-saved profiles are saved even if the base Essentium resources were replaced with newly updated Essentium files but NOT if a fresh Cura version to install was downloaded (See Figure 16).

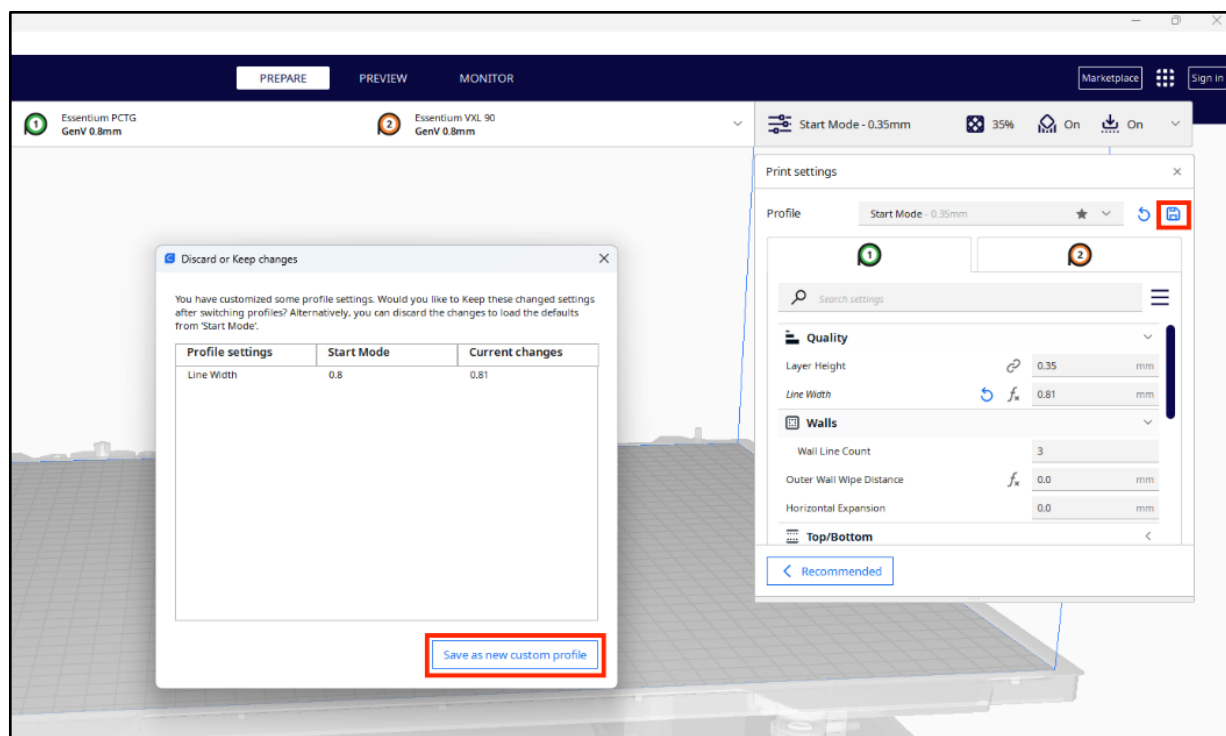


Figure 16: Saving a Custom Profile

INSTALLING AND SLICING – Mac Computer

This section shows the steps for installing and slicing Cura using a Mac Computer.

1. Select either of the two (2) software options to download based on which processing chip the computer used contains.
 - [MacOS-x64.dmg](#) – Intel® processor
 - [MacOS-ARM64.dmg](#) – Apple® M1 processor
2. Once downloaded, double-click on the *.dmg* file. A new screen appears. Drag and drop the UltiMaker Cura icon onto Applications folder icon (See Figure 17).



Figure 17: Accessing Cura

3. Once software installation is complete, open the UltiMaker Cura software (See Figure 18).

NOTE: Opening and immediately closing the software is necessary for performing future steps.

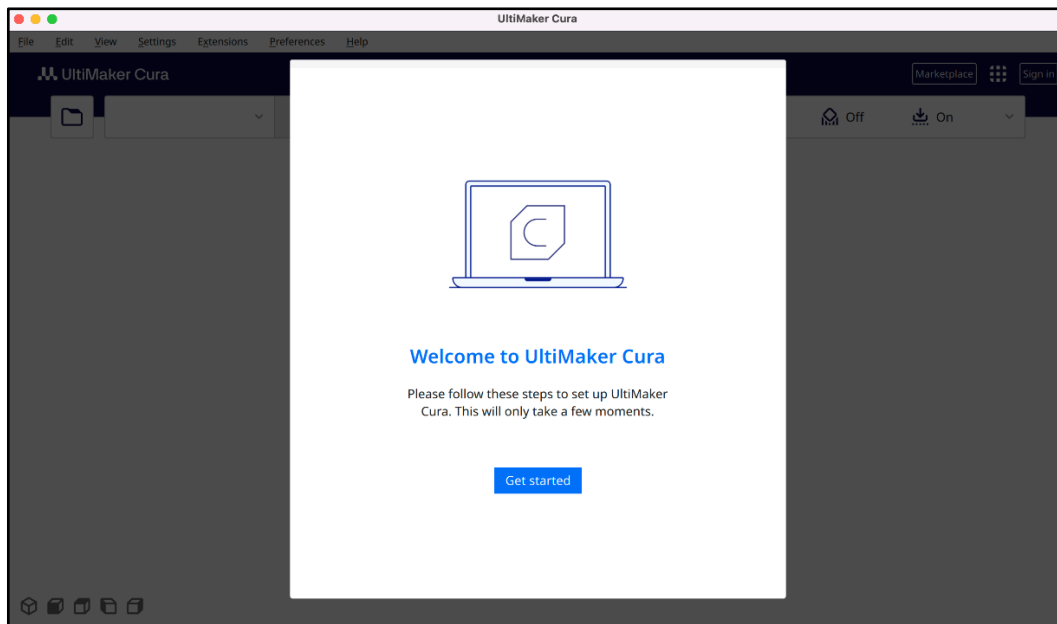


Figure 18: Cura Software Opening

4. Download the accompanying Essentium resources files (v2.0.0 zip file), and then unzip the file.
5. Keep File Explorer for this unzipped folder open on desktop.
6. Navigate to the Cura Configuration folder in File Explorer.

- Macintosh HD > Users > {user_name} > Library > Application Support > cura > 5.6
- **NOTE 1:** {user_name} will be specific to each user
- **NOTE 2:** “Library” folder is hidden. Within Finder press command+shift+.(period) to reveal hidden folders.

7. Copy and paste the Essentium resources files into the Cura Configuration folder “5.6” (See Figure 3). This folder will include the following subfolders:

- Definitions
- Extruders
- Intents (Currently empty placeholder)
- Materials
- Meshes
- Quality
- Scripts
- Setting_visibility
- Variants

There are 2 options for migrating the files from the downloaded .ZIP into the Cura Configuration folder. The first option may only be used if you have not saved any custom Cura profiles based on our Essentium materials. This includes saved profiles for all platforms (Essentium or other).

NOTE: If you bulk replace these folders and have custom saved profiles within Cura, you risk deleting your custom saved profiles.

Option 1 is simply selecting all the folders together and drag and dropping the folders into the 5.6 folder, as seen below:

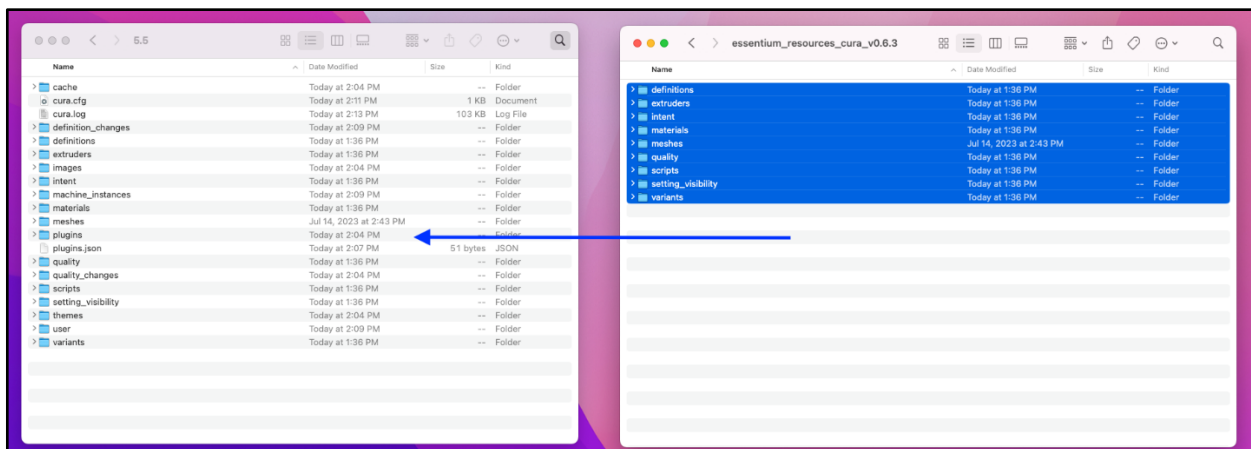


Figure 19: Copying and Pasting the Essentium Resource Files

Option 2 requires opening each individual folder above and dragging and dropping the contents directly into the corresponding folder within the 5.6 folder. This will need to happen for each folder listed above. This ensures that the files in this folder structure related to your custom saved profiles are not harmed during the file migration process.

8. Double-click on the UltiMaker Cura application within the Applications folder to reopen Cura. At this point, all Essentium machine and materials are available for selection and use.
9. The 'Getting Started' panel prompts the user to add the first machine into UltiMaker Cura (See Figure 20).

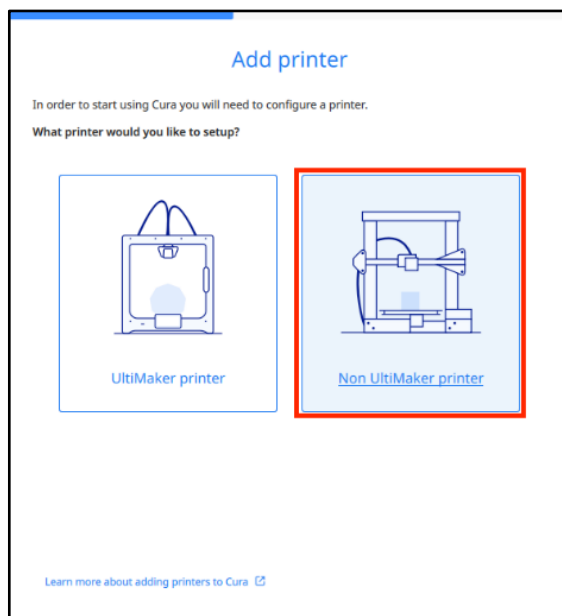
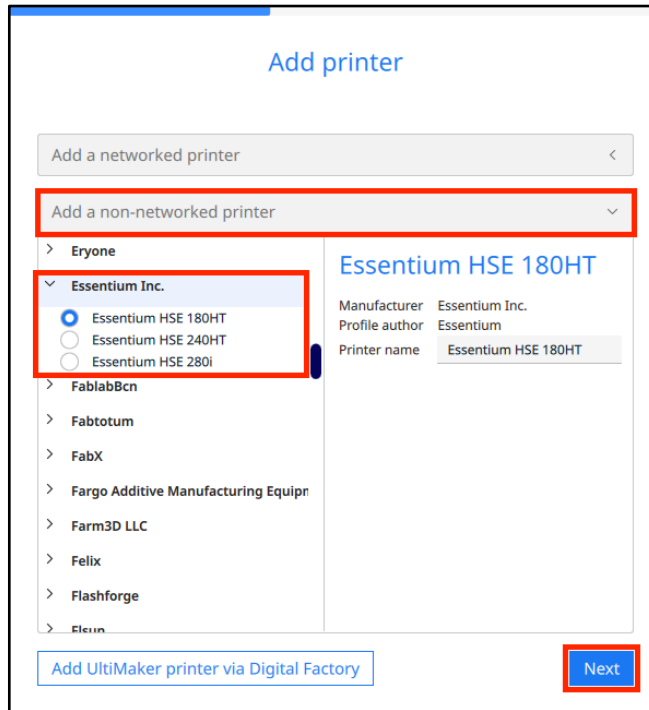


Figure 20: Adding a Printer

Essentium machines are listed under “Essentium, Inc” in the “non-networked” sub-section.

10. Select “Add a non-networked printer”

11. Select the desired HSE machine, and then select 'Next' (See Figure 21).



Add printer

Add a networked printer

Add a non-networked printer

> Eryone

▼ Essentium Inc.

- ☒ Essentium HSE 180HT
- ☐ Essentium HSE 240HT
- ☐ Essentium HSE 280i

> FablabBcn

> Fabtotum

> FabX

> Fargo Additive Manufacturing Equipm

> Farm3D LLC

> Felix

> Flashforge

> Fleum

Essentium HSE 180HT

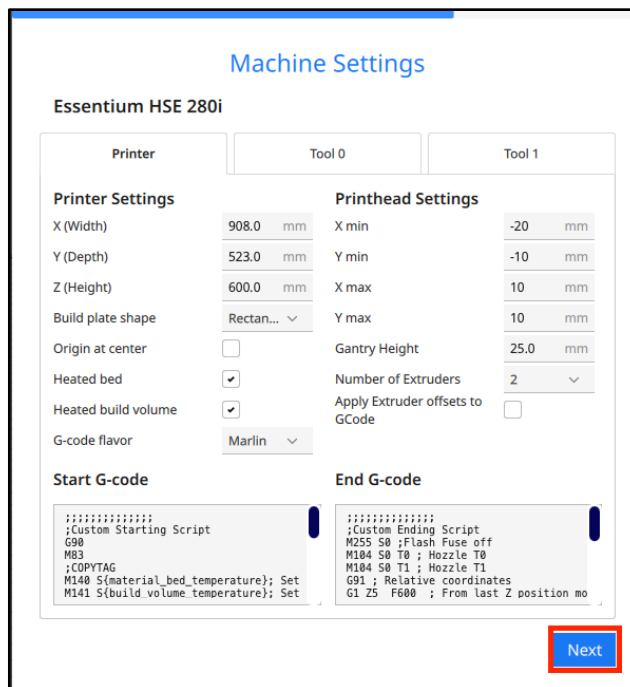
Manufacturer: Essentium Inc.
Profile author: Essentium
Printer name: Essentium HSE 180HT

Add Ultimaker printer via Digital Factory

Next

Figure 21: Adding an HSE Printer

12. Enter any desired settings on the 'Machine Settings' prompt, and then click 'Next' (See Figure 22).



Machine Settings

Essentium HSE 280i

Printer	Tool 0	Tool 1
Printer Settings		
X (Width)	908.0 mm	
Y (Depth)	523.0 mm	
Z (Height)	600.0 mm	
Build plate shape	Rectan...	
Origin at center	☐	
Heated bed	☒	
Heated build volume	☒	
G-code flavor	Marlin	
Start G-code		
<pre> ;Custom Starting Script G90 M83 ;COPYTAG M140 S{material_bed_temperature}; Set M141 S{build_volume_temperature}; Set </pre>		
Printhead Settings		
X min	-20 mm	
Y min	-10 mm	
X max	10 mm	
Y max	10 mm	
Gantry Height	25.0 mm	
Number of Extruders	2	
Apply Extruder offsets to GCode	☐	
End G-code		
<pre> ;Custom Ending Script M255 S0 ;Flash Fuse off M104 S0 T0 ; Nozzle T0 M104 S0 T1 ; Nozzle T1 G91 ; Relative coordinates G1 Z5 F600 ; From last Z position mo </pre>		

Next

Figure 22: Adding Machine Settings

13. Figure 23 shows where additional machines can be added at the upper-left corner of the screen.

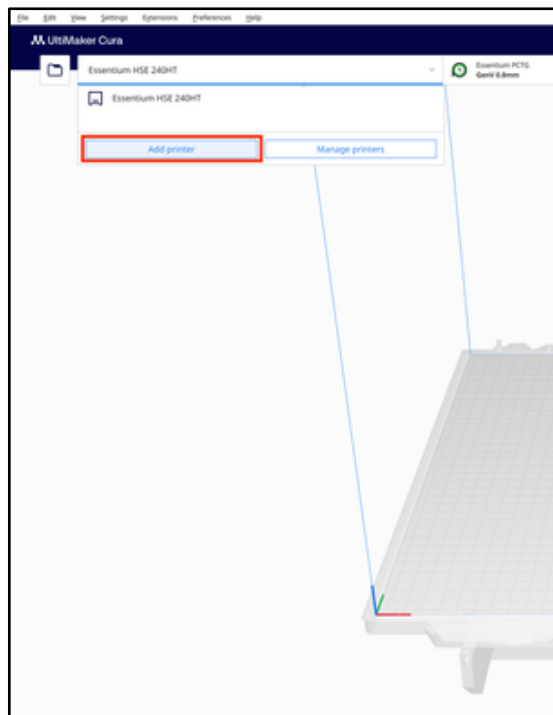


Figure 23: Adding Additional Printers

14. Navigate to Extensions > Post Processing > Modify G-Code (See Figure 24).

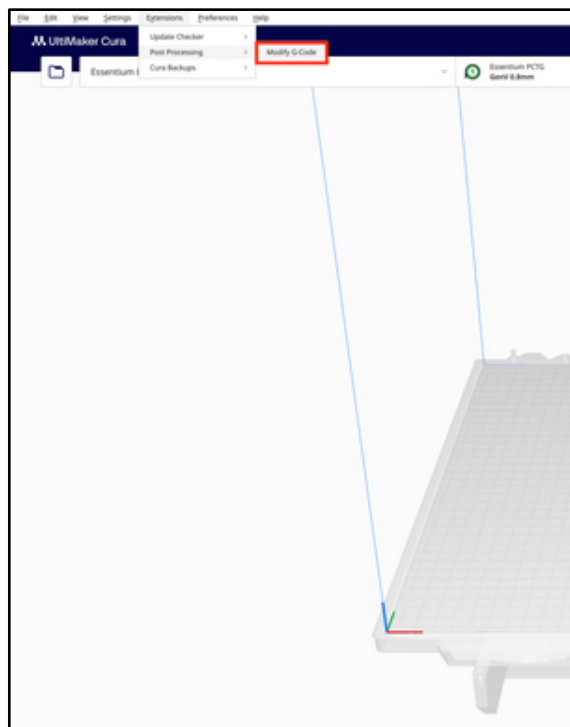


Figure 24: Navigating to Post-Processing Scripts

15. Click 'Add a script' (See Figure 25).

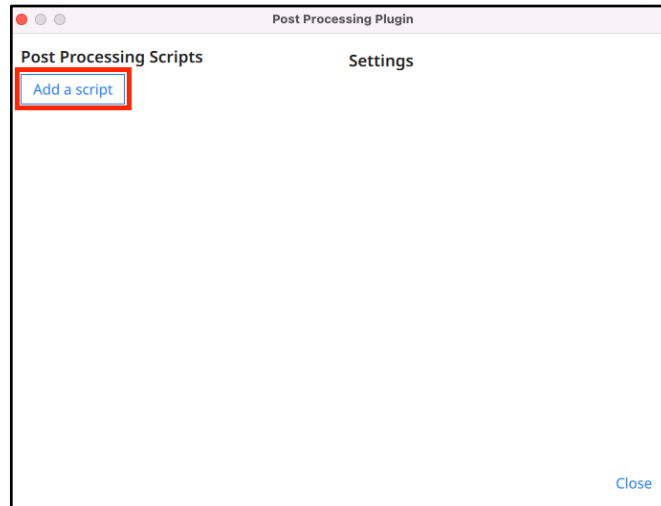


Figure 25: Adding a Script

16. Add the following post-processing scripts by selecting the name from the script, and then clicking 'Close' (See Figure 26):

- Essentium Post Processing Script
- Essentium Setting Injector



NOTE: This step must be done for each machine added.

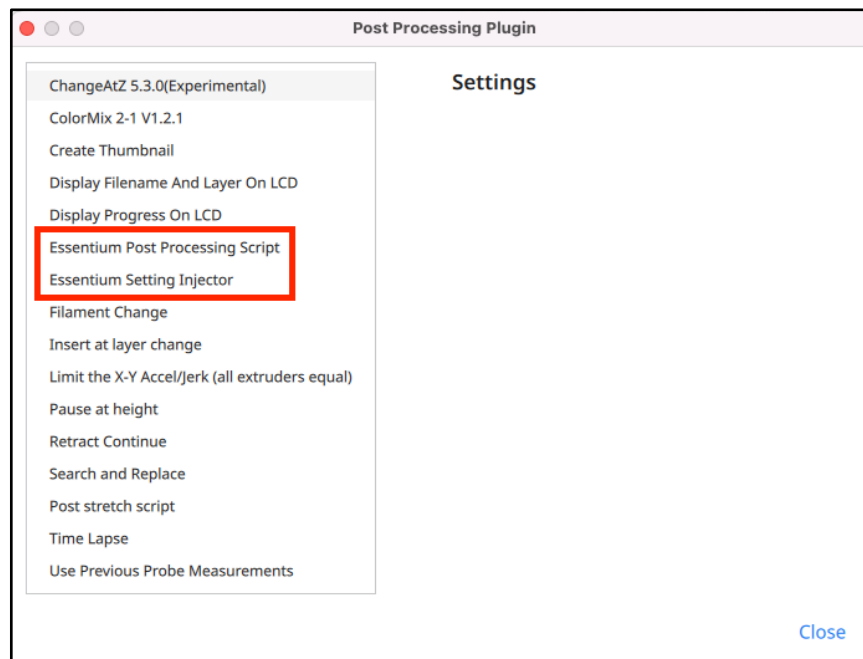


Figure 26: Adding Post-Processing Scripts

17. Select 'Marketplace' at the top-right of the screen, and then search for each plugin name, and then click 'Install' (See Figure 27).



Required plugin:

Linear Advance Settings

Optional plugins:

- Material Settings
- Printer Settings
- Settings Guide

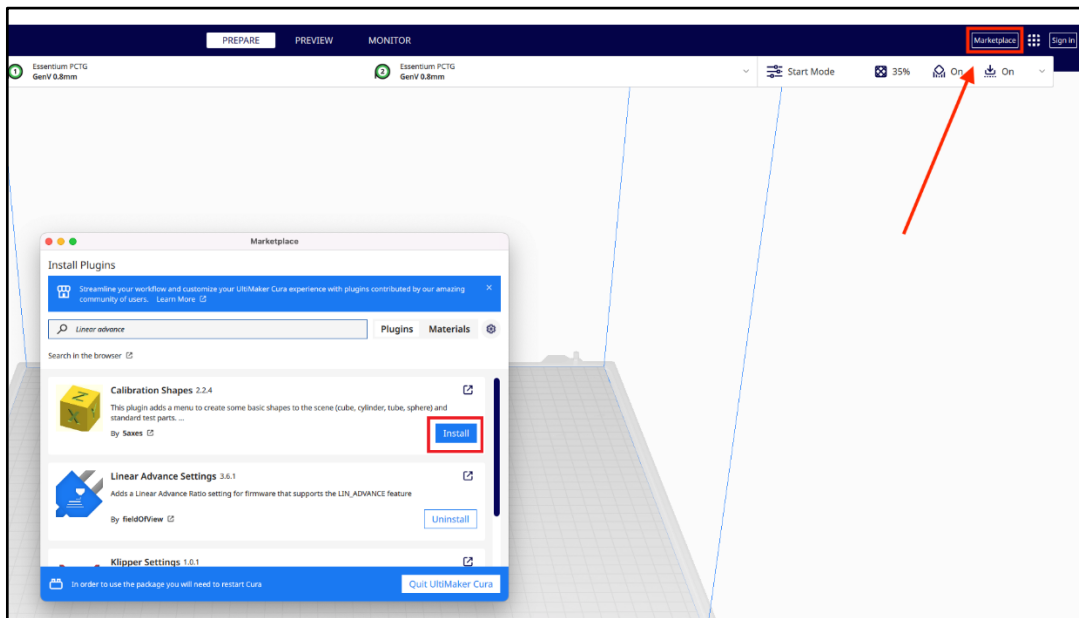


Figure 27: Adding Marketplace Plugins

18. Restart the Ultimaker Cura software for plugins to take effect.

At this point, the Ultimaker Cura software is ready for printing. The steps below show a quick overview for importing a part, .stl (or .obj), and for selecting the user's materials and hotend of choice.

NOTE: At this stage, a majority of the application-specific profiles are not different from "Start Mode".

Essentium has a custom option for which settings are visible at the right-hand panel showing the settings that are most often modified in Cura for HSE printers (See Figure 28).

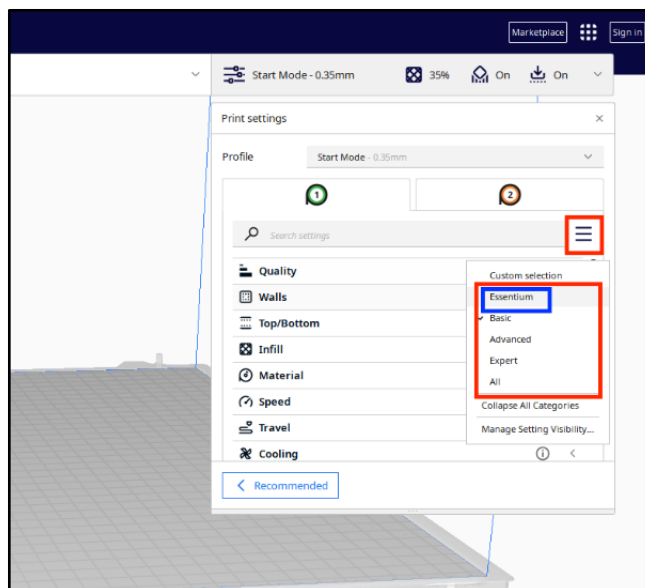


Figure 28: Modifying Settings

19. Restart the UltiMaker Cura software for plugins to take effect.

At this point, the UltiMaker Cura software is ready for printing. The steps below show a quick overview for importing a part, *.stl* (or *.obj*), and for selecting the user's materials and hotend of choice.

NOTE: At this stage, a majority of the application-specific profiles are not different from "Start Mode".

20. Select the desired material and hotend/nozzle (See Figure 29).

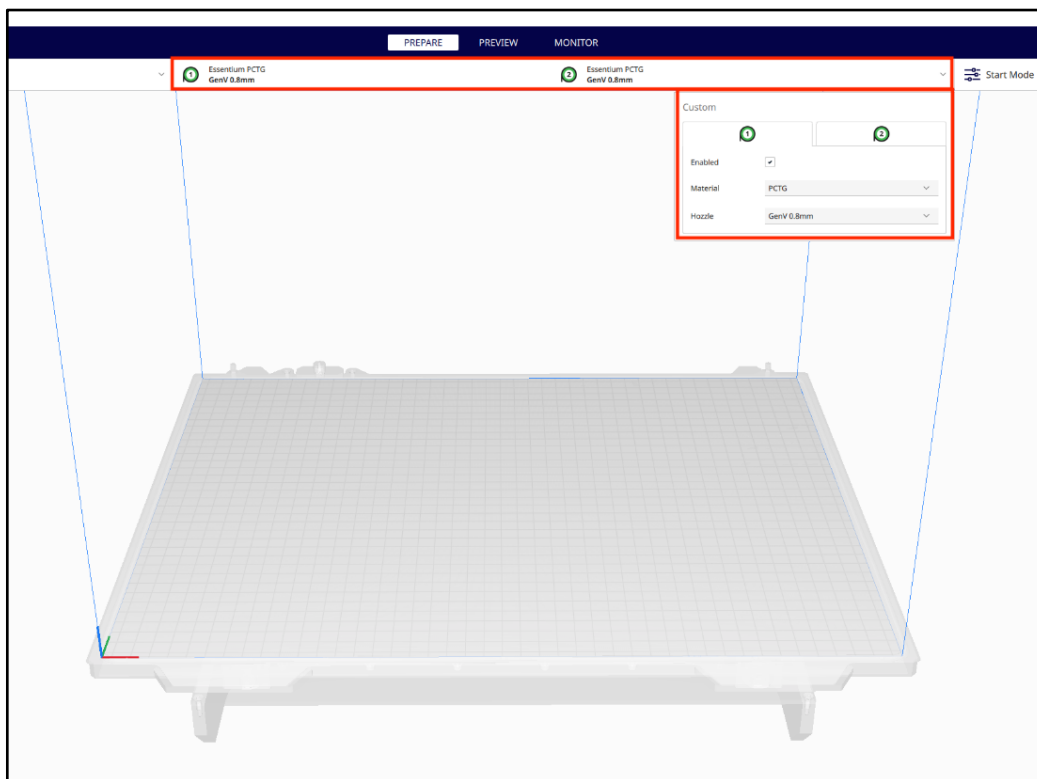


Figure 29: Selecting Material and Hotend/Nozzle

21. Select settings visibility. Essentium has a custom option for which settings are visible at the right-hand panel, showing the settings that are most often modified in Cura for HSE printers (See Figure 30).

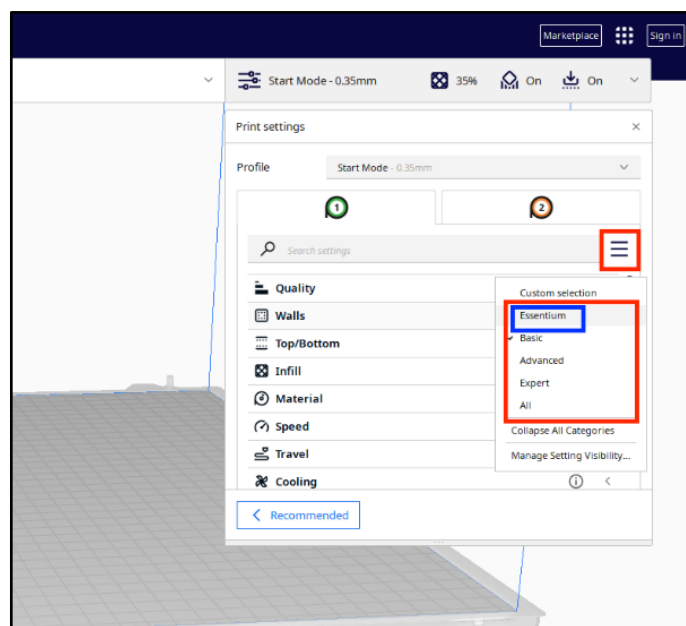


Figure 30: Modifying Settings

22. Click the folder icon in the upper-left corner of the page to select a part (See Figure 14).

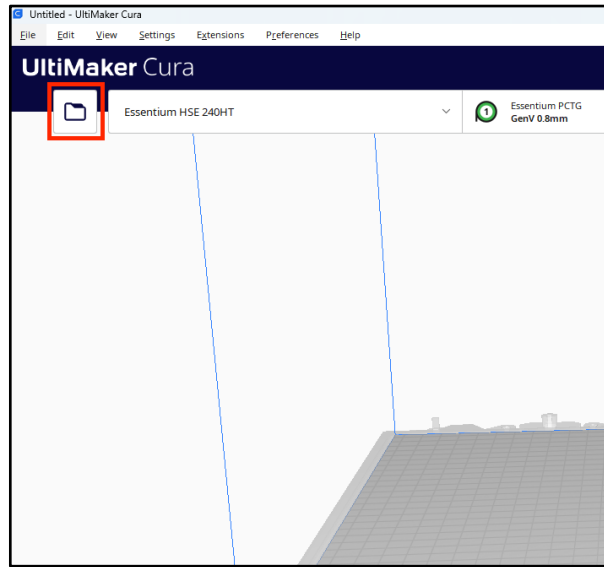


Figure 31: Selecting a Part

23. Click on 'Slice' in lower-right hand corner of the page to begin slicing (See Figure 32).



NOTE: Upon adding a part to the environment, an icon with a red “2” appears next to the 'Slice' button. This is an indication that the two (2) post-processing scripts enabled earlier in this procedure are active. Verify that this icon appears next to the 'Slice' button when printing on HSE machines, as the print will not execute without these post-processing scripts enabled.

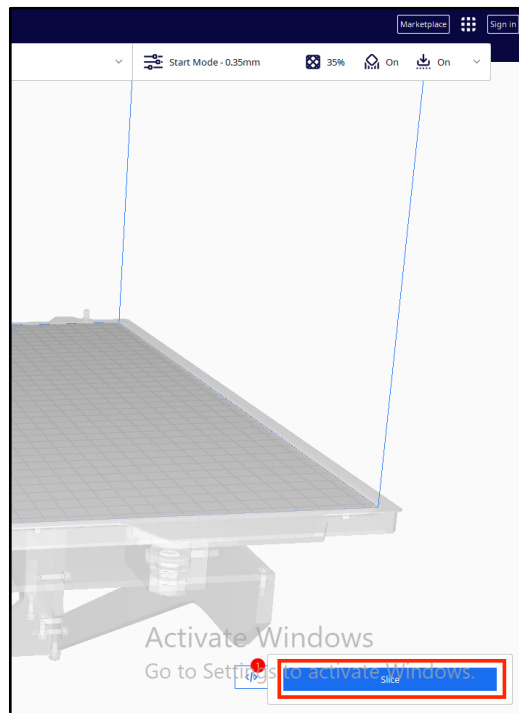


Figure 32: Slicing

At this point, the user can view the toolpath created for that specific part as well as save the .gcode.

24. (Optional Step) The 'Save' button only shows up once a modification is made from the base settings.

NOTE: These user-saved profiles are saved even if the base Essentium resources were replaced with newly updated Essentium files but NOT if a fresh Cura version to install was downloaded (See Figure 33).

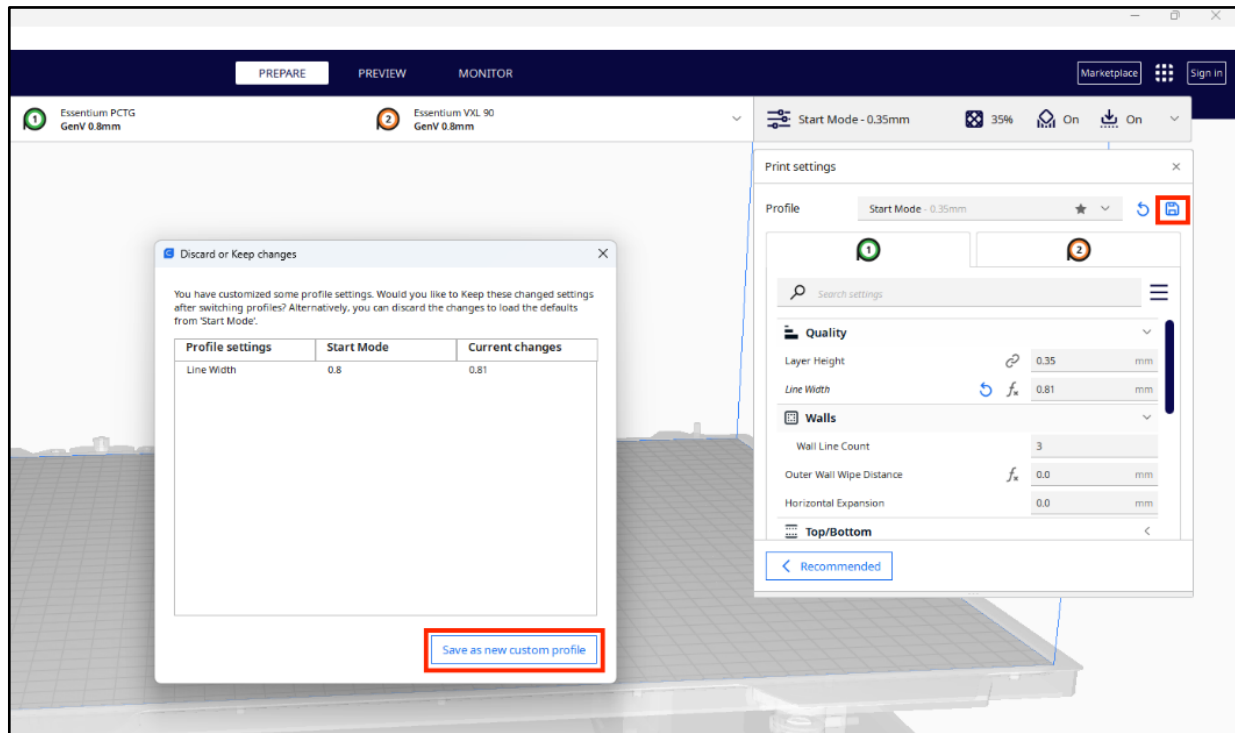


Figure 33: Saving a New Custom Profile

25. (Optional Step) The 'Save' button only shows up once a modification is made from the base settings.

NOTE: These user-saved profiles are saved even if the base Essentium resources were replaced with newly updated Essentium files but NOT if a fresh Cura version to install was downloaded (See Figure 34).

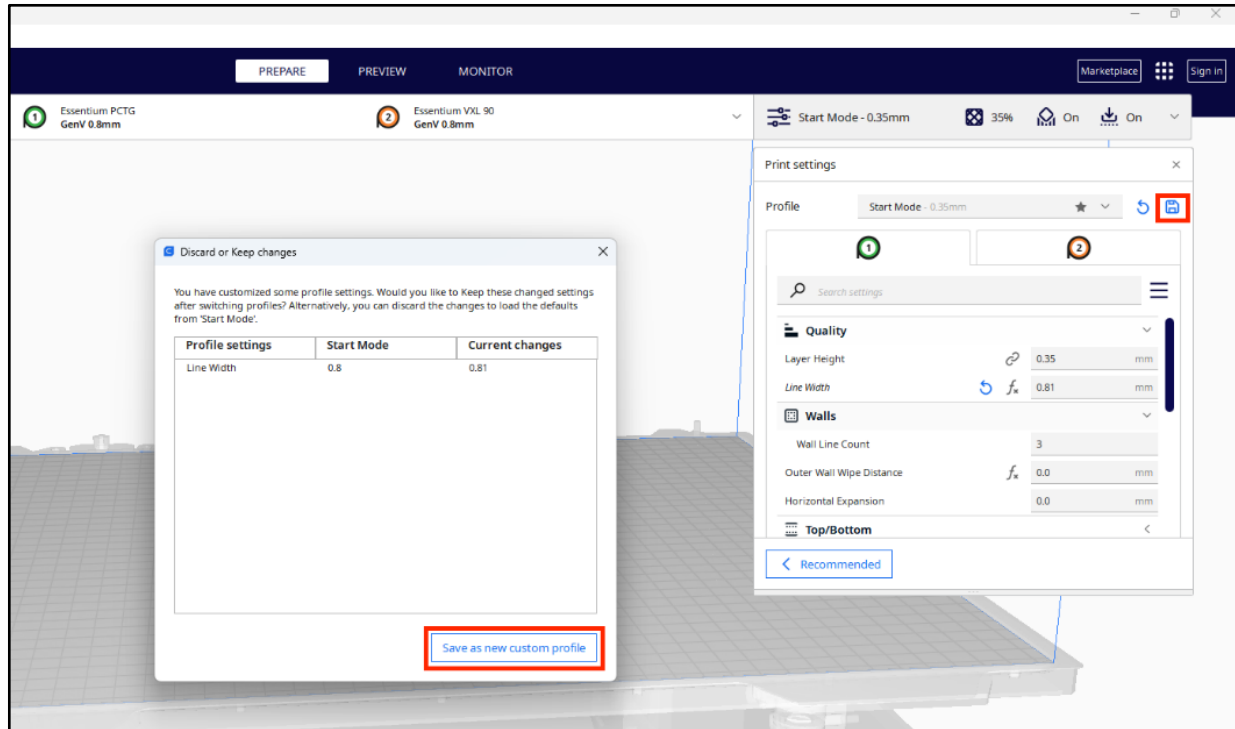


Figure 34: Saving a Custom Profile

UPDATING CURA WITH NEW ESSENTIUM RESOURCES – PC OR MAC

When Essentium releases new or updated print profiles a similar installation/migration procedure will need to occur.

Please repeat steps 4-7 from above steps (PC or MAC) now with the new Essentium resources .zip. There is a similar alternative to navigating to the Cura Configuration folder in Step 6. With Cura open, navigate to Help -> Show Configuration Folder on the top navigation bar. The opened folder is the specific location to add/replace new print profiles.

Please pay special attention to the 2 migration options in step 7. This is a concern if you have previously saved Cura profiles within Cura.

SUPPORT AND HOTEND/NOZZLE COMPATABILITY

Table 1 shows compatibility for support material and the hotend/nozzle.

Table 1: Support and Hotend/Nozzle Compatibility

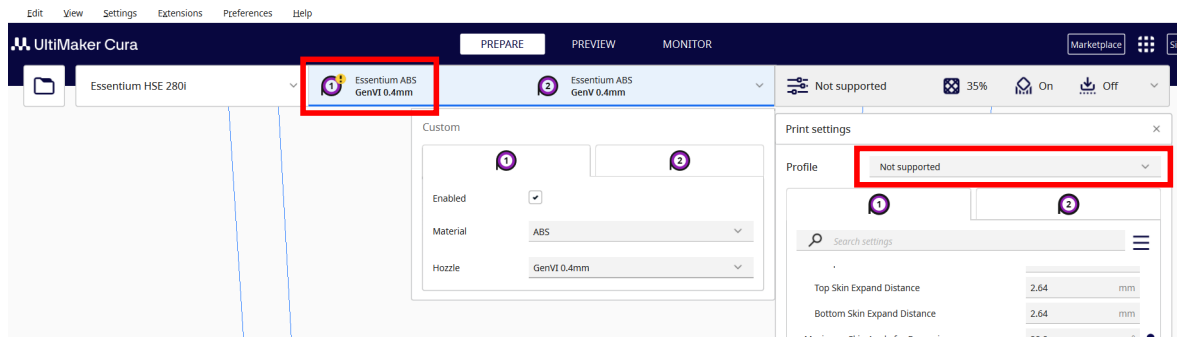
Model	Support	Hotend/Nozzle
PCTG	VXL 90, PLA	GenV Hotend
PETCF	VXL 90, PLA	GenV Hotend
HTN	VXL 90 or S10	GenV Hotend
HTNCF	VXL 90 or S10	GenV Hotend
PACF	VXL 90 or S10	GenV Hotend
TPU-74D	VXL 90 or S10	GenV Hotend
TPU-95A	VXL 90 or S10	GenVI Hotend
ABS	VXL 90, HIPS, PCTG	GenV Hotend
9085	3DX Tech, B20	GenV Hotend
Duratem	PC	GenV Hotend
PEEK	Native only	GenVI Hotend
PEKK	Native only	GenV Hotend
PPSCF	Native only	GenV Hotend
Altitude	Native only	GenV Hotend



IMPORTANT NOTES

The following are important usage notes, known issues, and common errors.

- Not all combinations of printers, hotends, and materials are currently supported. Do not attempt to slice or print any gcode generated if a toolhead has a yellow (!) icon or if the Print settings Profile indicates “Not supported.”



The most common cause of an unsupported configuration error comes from attempting to use a GenVI hotend with a material profile which has not yet been officially released. Please contact support for instructions on using a GenVI hotend with material for which a profile has not yet been officially released.

- Using and enabling the Essentium post processing scripts is critical for proper functionality of the HSE printer. The Essentium post processing scripts convert the default, Cura generated g-code, into g-code which is compatible with the HSE printers. Please be sure these are selected and enabled for each HSE configured within Cura.
- There is currently a known compatibility issue with Cura’s “Z Hop When Retracted” setting and the HSE printers. Please do NOT enable this setting.